



Elastic Strings and Springs · Worksheet 1

Further Mechanics 1 Chapter 3: Elastic Strings and Springs. Answer all questions. Show your working clearly. Total: 47 marks. Time: 60 minutes.

Name _____

Class _____

Date _____

INSTRUCTIONS

- Answer **all** questions in the spaces provided. Show all your working.
- There are two sections, one for each topic in Chapter 3. Within each section the questions increase in difficulty; the **last question** of each section is set at A level exam standard.
- Take $g = 9.8 \text{ m s}^{-2}$. Use $T = \frac{\lambda x}{l}$ and elastic potential energy $\frac{\lambda x^2}{2l}$. An elastic string can only pull; a spring can pull or push. Give non-exact answers to 3 significant figures.
- The **Roar-level challenge** at the end is a harder stretch question.
- The marks for each question are shown in brackets, e.g. [3].

Section	Marks available	Marks scored
A Hooke's law and equilibrium	20	
B Elastic potential energy and energy problems	22	
Roar-level challenge	5	
Total	47	

Section A Hooke's law and equilibrium

20 marks

1. A light elastic string has natural length 0.8 m and modulus of elasticity 20 N. It is stretched to a length of 1 m. [2]
Find the tension in the string.

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2. A particle of mass 2 kg hangs in equilibrium from one end of a light elastic string of natural length 1.2 m. The other end of the string is attached to a fixed point. The extension of the string is 0.3 m. [3]
Find the modulus of elasticity of the string.

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3. A light spring of natural length 0.6 m and modulus of elasticity 49 N stands vertically with its lower end fixed to the ground. A particle of mass 1.5 kg rests in equilibrium on top of the spring. Find the length of the spring. [3]

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4. A particle of mass m kg hangs in equilibrium from a light elastic string of natural length 0.5 m and modulus of elasticity λ N, whose other end is fixed. The extension of the string is 0.1 m. When a second particle of mass 0.4 kg is attached to the first, the new equilibrium extension is 0.15 m. Find λ and m . [4]

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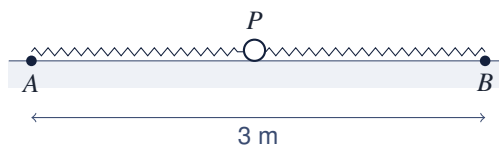
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5. Two fixed points A and B are 3 m apart on a smooth horizontal surface. A particle P is attached to one end of each of two light elastic strings. The other end of the first string is attached to A ; this string has natural length 1 m and modulus of elasticity 20 N. The other end of the second string is attached to B ; this string has natural length 1.2 m and modulus of elasticity 30 N. P rests in equilibrium on the line AB .



- (a) Find the distance AP . [4]

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- (b) Find the tension in each string. [2]

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- (c) P is moved along AB to the midpoint of AB and held at rest. Find the magnitude and direction of the resultant force exerted on P by the strings. [2]

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(Total for Question 5 is 8 marks)

**Section B Elastic potential energy and energy problems**

22 marks

6. A light elastic string has natural length 1.5 m and modulus of elasticity 30 N. [2]

Find the elastic potential energy stored in the string when its extension is 0.5 m.

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7. A light elastic string has natural length 0.8 m and modulus of elasticity 40 N. [3]

Find the work done in stretching the string from an extension of 0.2 m to an extension of 0.6 m.

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8. A light spring of natural length 0.4 m and modulus of elasticity 20 N lies on a smooth horizontal surface with one end fixed. A particle of mass 0.5 kg is held against the free end so that the spring is [3]

compressed by 0.1 m. The particle is released from rest.

Find the speed of the particle when the spring reaches its natural length.

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9. A particle of mass 0.5 kg is attached to one end of a light elastic string of natural length 1 m and modulus of elasticity 19.6 N. The other end of the string is attached to a fixed point O . The particle is [4]

held at O and released from rest, and falls vertically.

Find the greatest distance below O reached by the particle.

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10. A bungee jumper of mass 70 kg is attached to one end of a light elastic rope of natural length 20 m and modulus of elasticity 1680 N. The other end of the rope is fixed to a point O on a high bridge. The jumper steps off the bridge at O and falls vertically from rest. The jumper is modelled as a particle and air resistance is ignored.

(a) Find the greatest distance below O reached by the jumper. [4]

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(b) Find the speed of the jumper at the point 30 m below O . [2]

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(c) By considering the point at which the acceleration of the jumper is zero, find the maximum speed of the jumper. [4]

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(Total for Question 10 is 10 marks)



ROAR-LEVEL CHALLENGE

5 marks

11. A particle P of mass 2 kg is attached to one end of a light elastic string of natural length 1.5 m and modulus of elasticity 36 N. The other end of the string is fixed to a point A on a rough plane inclined at 30° to the horizontal. The coefficient of friction between P and the plane is 0.25. P is held at A and released from rest, and slides down a line of greatest slope. Find the greatest extension of the string. [5]

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END OF WORKSHEET

Total: 47 marks